# Practice workbook · Write a short number program from a list of requirements

From a bullet list of requirements (no design given) the student writes a short numeric program: inputs converted to numbers, calculations with arithmetic/modulus operators, selection for bands or tests, and count- or condition-controlled loops, including stepping through an array.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-05

# 1. Question 3 · part whole · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

**3** Manjit sells copies of a science textbook to schools.

She needs a program to process textbook orders. It must:

- accept the input of the number of textbooks required
- generate the price per textbook
- generate the total cost of the order
- display the price per textbook and the total cost of the order.

The price of one textbook depends on the number ordered:

| Quantity | Price per textbook |
|---|---|
| 1–5 | £20.00 |
| 6–9 | £15.00 |
| 10 or more | £12.00 |

Open **Q03** in the code editor.

Write the program.

You must use the structure given in **Q03** to complete the program.

Do not add any further functionality.

Save your code as **Q03FINISHED** with the correct file extension for the programming language.

**(Total for Question 3 = 6 marks)**

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## 1. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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## 2. Question 3 · part c · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

**3** Different algorithms can be used to manipulate data.

(a) **Email.txt** contains a list of email addresses.

Open **Q03a** in the code editor.

Write a program to implement these requirements.

The code must:

- check each email address to ensure it contains the '@' symbol.
- write email addresses that do not contain the '@' symbol to an **Error.txt** file.

You must use the structure given in the file **Q03a** to complete the program.

Do not add further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

**(6)**

(b) Explain **one** drawback of using the merge sort algorithm to sort large data sets.

**(2)**

(c) Open **Q03c** in the code editor.

Write a program to display the square and cube of a number between 1 and 50 entered by a user.

The code must:

- ask the user to enter a number between 1 and 50 inclusive
- display the number, the square of the number and the cube of the number, with appropriate labels
- stop when a number outside the range 1 to 50 is entered.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

**(6)**

**(Total for Question 3 = 14 marks)**

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## 2. Question 3 · part c · 6 marks · Working and answer

Working and answer

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## 3. Question 2 · part b · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

**2** Rama wants some computer programs for his daughter, Ayomi.

(a) Rama wants a program to calculate the area of a triangle.

(area of a triangle = ½ base x height)

This pseudocode contains the logic required to create the program.

```
1 #  Initialise variables
2
3 SET base TO 50
4 SET heightChk TO TRUE
5
6 #  Print prompt and take input from user
7
8 WHILE heightChk DO
9     SEND "Enter the height (between 1 and 100): " TO DISPLAY
10    RECEIVE height FROM (INTEGER) KEYBOARD
11    IF (height >= 1 AND height <= 100)
12    THEN heightChk = FALSE
13    END IF
14 END WHILE
15
16 #  Calculate and print out values
17
18 SET area TO 0.5 * base * height
19
20 SEND ("Base of triangle is: " , base) TO DISPLAY
21 SEND ("Height of triangle is: " , height) TO DISPLAY
22 SEND ("Area of triangle is: " , area) TO DISPLAY
```

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# 3. Question 2 · part b · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

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Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

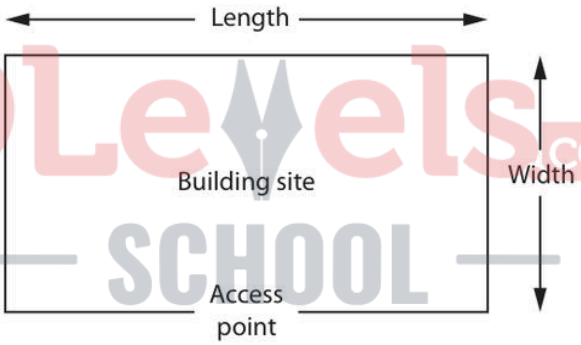
You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

(b) Ayomi wants a program that will calculate the number of fence panels required to surround building sites such as the one shown in the diagram (not to scale).




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Each building site is a rectangle.

Each fence panel is 1 metre long.

A gap of 4 metres is needed to access each site. The location of the gap is not significant.

The user will input the length and width of the building site as whole numbers of metres.

The program will output the minimum number of full panels needed.

**Q02b** provides a structure for the program.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to write the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(7)

**(Total for Question 2 = 17 marks)**

## 3. Question 2 · part b · 7 marks · Working and answer

Working and answer

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## 4. Question 1 · part e · 5 marks

2021 NOV · Paper 02

Further practice · Attempt part e only.

**1** Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

| Error description | Logic | Syntax | Runtime |
|---|---|---|---|
| Divide by 0 | | | |
| Use x instead of * to multiply | | | |
| Subtract 10 from 2 instead of 2 from 10 | | | |

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# 4. Question 1 · part e · 5 marks

2021 NOV · Paper 02

Further practice · Attempt part e only.

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(c) **Figure 1** shows a programming construct in a flowchart.

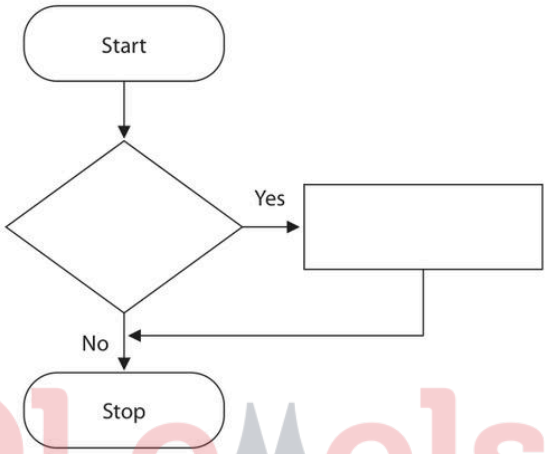



**Figure 1**

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

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# 4. Question 1 · part e · 5 marks

2021 NOV · Paper 02

Further practice · Attempt part e only.

(e) A program is needed to calculate and print a multiplication table for a number that is input by the user.

**Figure 2** shows the display when the number 2 is input.

```
Enter the number: 2
1 x 2 is 2
2 x 2 is 4
3 x 2 is 6
4 x 2 is 8
5 x 2 is 10
6 x 2 is 12
7 x 2 is 14
8 x 2 is 16
9 x 2 is 18
10 x 2 is 20
11 x 2 is 22
12 x 2 is 24
```

**Figure 2**

Open **Q01e** in the code editor.

Write the program.

You **must** use the structure given in the file **Q01e** to complete the program.

Do not add any further functionality.

Save your code as **Q01eFINISHED** with the correct file extension for the programming language.

**(5)**

**(Total for Question 1 = 13 marks)**

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## 4. Question 1 · part e · 5 marks · Working and answer

Working and answer

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## 5. Question 3 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

**3** Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

| Type of test data | Test data |
|---|---|
| Normal | 3 |
| | 1 |
| | 15 |

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## 5. Question 3 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

(b) Explain **one** reason why parameters are used to pass data into subprograms.

(2)

(c) A leap year is a year that has 366 days rather than 365 days. A leap year usually occurs every four years.

A year that is a leap year:

- can be exactly divided by 400

Or

- can be exactly divided by 4, but **not** exactly divided by 100.

These years are leap years: 2000, 2012 and 2020.

These years are not leap years: 1700, 1900 and 2021.

A program is needed that will accept the input of a year and output whether it is a leap year or not.

Open **Q03c** in the code editor.

Write the program.

You **must** use the structure given in **Q03c**.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

**(Total for Question 3 = 15 marks)**

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## 5. Question 3 · part c · 6 marks · Working and answer

Working and answer

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## 6. Question 3 · part b · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

**3** Programs use logic, display data and validate input.

(a) The truth table shows the initial values for A and B.

Complete the truth table to show the results of each operation.

(4)

| A | B | A OR B | NOT B | (A OR B) AND (NOT B) |
|---|---|---|---|---|
| 0 | 0 | | | |
| 0 | 1 | | | |
| 1 | 0 | | | |
| 1 | 1 | | | |

(b) **Figure 2** shows the intended output from a program that displays every other name in an array of star names.

```
Alasia
Castor
Electra
Gudja
Izar
Kang
Maia
Ogma
```

**Figure 2**

Open **Q03b** in the code editor.

Amend the code to display every other name in the array of star names.

Do not add any further functionality.

Save your code as **Q03bFINISHED** with the correct file extension for the programming language.

(3)

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## 6. Question 3 · part b · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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(c) A program validates a number input by the user.

**Figure 3** shows the output messages based on the inputted number.

| Input | Output |
|---|---|
| <empty> | You must provide a number |
| Any negative number | The number must be greater than zero |
| 0 | The number must be greater than zero |
| 1 to 20 | Acceptable |
| 60 or more | Acceptable |
| 31 to 39 | Centre |
| 30 | Perfect |
| Any other number | No message |

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**Figure 3**

Open **Q03c** in the code editor.

Amend the code to ensure the messages are generated correctly.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

**(Total for Question 3 = 13 marks)**

## 6. Question 3 · part b · 3 marks · Working and answer

Working and answer

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